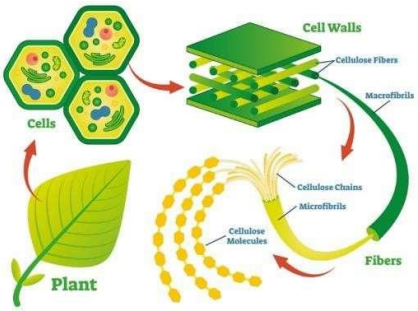
Module – 2 : APPLICATION OF BIOMOLECULES

Carbohydrates in cellulose-based water filters production, PHA and PLA in bioplastics production, Nucleic acids in vaccines and diagnosis, Proteins in food production, lipids in biodiesel and detergents production, Enzymes in biosensors fabrication, food processing, detergent formulation and textile processing.

Biomolecule, also called biological molecule: is any of numerous substances that are produced by cells and living organisms. The four major types of biomolecules are carbohydrates, lipids, nucleic acids, and proteins.

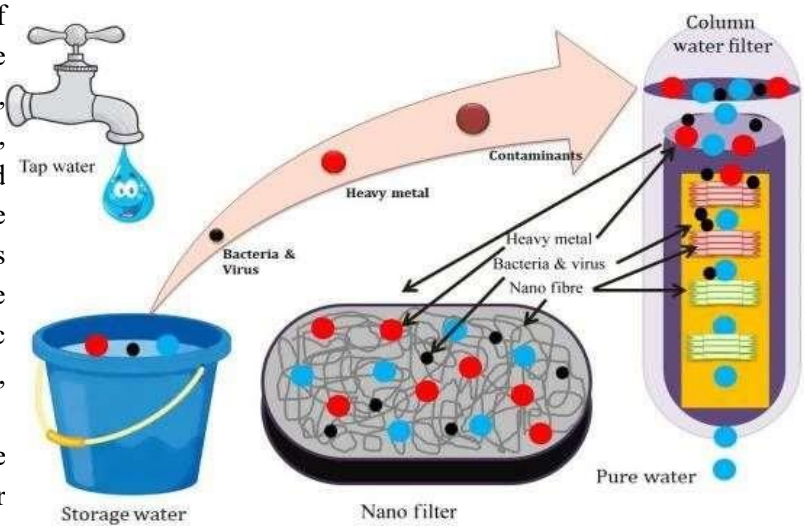
Carbohydrates in Cellulose-Based Water Filters Production

Cellulose



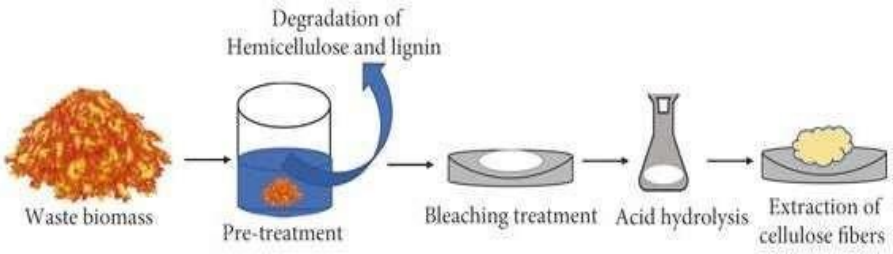
- Cellulose is a complex carbohydrate consisting of Oxygen, Carbon, and Hydrogen with a molecular formula of $(C_6H_{10}O_5)_n$
- Cellulose is the main structural molecule of cell walls of plants and algae.
- Cellulose is Chiral, tasteless and odourless.
- Cellulose comprises about 33 percent of all vegetable matter. Wherein, 90 percent of cotton and 50 percent of wood are cellulose.

Cellulose-Based Water Filters Production



- Water filtration is the process of removing or reducing the concentration of particulate matter, including suspended particles, parasites, bacteria, algae, viruses, and fungi, as well as other undesirable chemical and biological contaminants from contaminated water to produce safe and clean water for a specific purpose, such as drinking, medical, and pharmaceutical applications.
- Cellulose-based filters are the affordable and biobased option for water filtration.
- Filters based on cellulose pulp fibres, usually have large pores that facilitate water percolation but they do not remove bacteria due to large pore size.
- Using mechanical or chemical treatment, cellulosic materials can be converted into cellulose nanofibers and nanocrystals which have excellent capabilities in water purification applications
- Cellulose fibres in two types of commercial paper filters have been given a positive net charge to trap bacteria through electrostatic interactions without releasing any biocides.
- Both Gram-positive and Gram-negative bacteria have a negative net surface charge on the cell envelope, due to peptidoglycans, liposaccharides and proteins in the cell wall, and this makes their removal non-selective and efficient for most types of bacteria.

Extraction of Cellulose and Preparation of cellulose based water filter:



- Cellulose containing materials such as wood, initially undergoes mechanical pre- treatment such as Cryocrushing and powdering, followed by treatment with Alkali/Sodium hydroxide, followed by delignification.
- This is followed by bio bleaching, which involves removal of residual ink by treatment with the enzymes.
- The bleached biomass is subjected to acid hydrolysis (Treating with Sulphuric acid), to obtain the cellulose fibres.
- The pre-treated cellulosic fibres are either subjected to homogenization/Ultrasonication/micro fluidization to obtain nanocellulose
- Cellulose fibres can be made into the nanofibers by electrospinning.
- The electrospinning technique provides the mat like structure of cellulose nanofibers twisted to prepare membrane.
- Several layers of the membrane are placed one above the other, which can be used as a filter paper.
- The cellulose-based filters can also be incorporated with antibacterial metal nanoparticles such as silver nanoparticles (AgNPs) and Copper nanoparticles (CuNPs) which are known to have antibacterial properties.

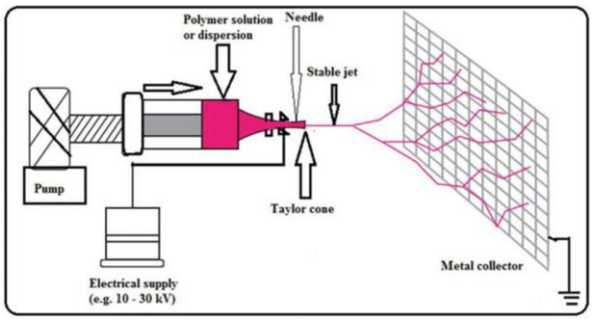
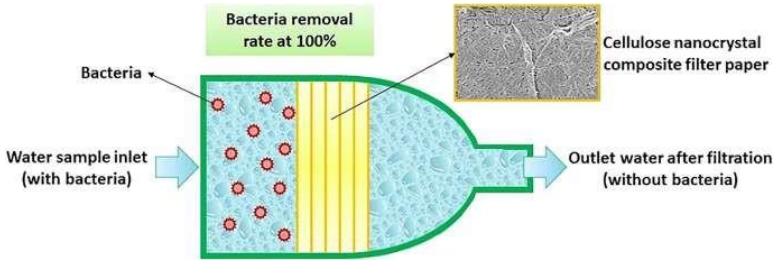


Fig. Electrospinning technique

The main advantages of Cellulose-Based Water Filters are :

- Economical, environmentally friendly disposal
- Harmless from an occupational medicine point of view: Organic Reliable and highly effective.
- Reduced consumption: less waste, lower disposal costs, less product loss, efficient and economic filtration
- If the filter sludge is compressed, cellulose filter aids can lead to-Higher yields of recovered residual filtrate which is more stable compacted material or briquettes which can, in part, be profitably recycled with less wear on the presses used for briquetting.



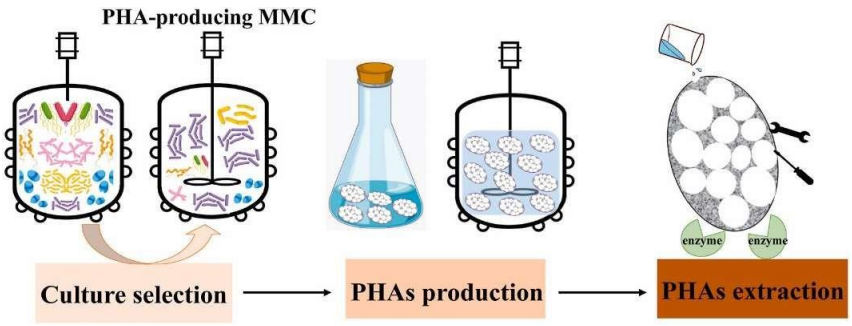
BIOPLASTICS

- Bioplastics are one type of plastic which can be generated from natural resources such as starches and vegetable oils.
- Bioplastics are basically classified as bio based and/or biodegradable. Not all bio-based plastics are biodegradable and similarly not all biodegradable plastics are bio based.
- Bio plastics are said to be biodegradable if they are broken down with the effect of the right environmental conditions and microbes which in turn use them as a food source.
- The bioplastics are considered compostable if within 180 days, a complete microbial assimilation of the fragmented food source takes place in a compost environment.
- Biobased plastics are made from a wide range of renewable Biobased feed stocks such as
  1. Agriculture based feedstocks: Plants rich in carbohydrates such as corn or sugarcane
  2. Lignocellulosic feedstocks: Plants that are not eligible for food or feed
  3. Organic waste feed stocks.

The two main types of Bioplastics are PHA (Poly Hyroxyalkonates) and PLA (Poly Lactic acid)

1) PHA (Poly Hydroxyalknoate):

- PHAs are polyesters synthesized and stored by various bacteria and archaea in their cytoplasm as water-insoluble inclusions.
- These are the Polyesters produced in nature by numerous microorganisms, such as bacterial fermentation of sugars or lipids.
- The PHA can weigh as much as 80 % of the organism's dry weight.
- The biosynthesis of PHA is usually caused by certain deficient conditions such as the lack of macro elements (phosphorus, nitrogen, trace elements, oxygen) and the excess supply of carbon sources.
- PHA production using waste plant oils (e.g., palm kernel oil, crude palm oil, palm oil, jatropha oil, sludge palm oil, soybean oil) as inexpensive carbon sources has been taken into consideration owing to its widespread availability and renewability.
- They are the Potential replacement for some petrochemical-based plastics.
- The raw materials that are mostly commonly used for PHA production are food crops, sugarcane, and vegetable oil.
- The PHA is biodegradable by soil, salt water, fresh water, aerobic industrial compost, anaerobic industrial compost, home compost typically in 6-8 weeks.
- The biosynthesis of PHA is usually caused by certain deficient conditions such as the lack of macro elements (phosphorus, nitrogen, trace elements, oxygen) and the excess supply of carbon sources.
- PHA recovery is an important factor which influences the total cost of the production process.
- PHAs are produced intracellularly, so they must be extracted using different solvents



Properties of PHA:

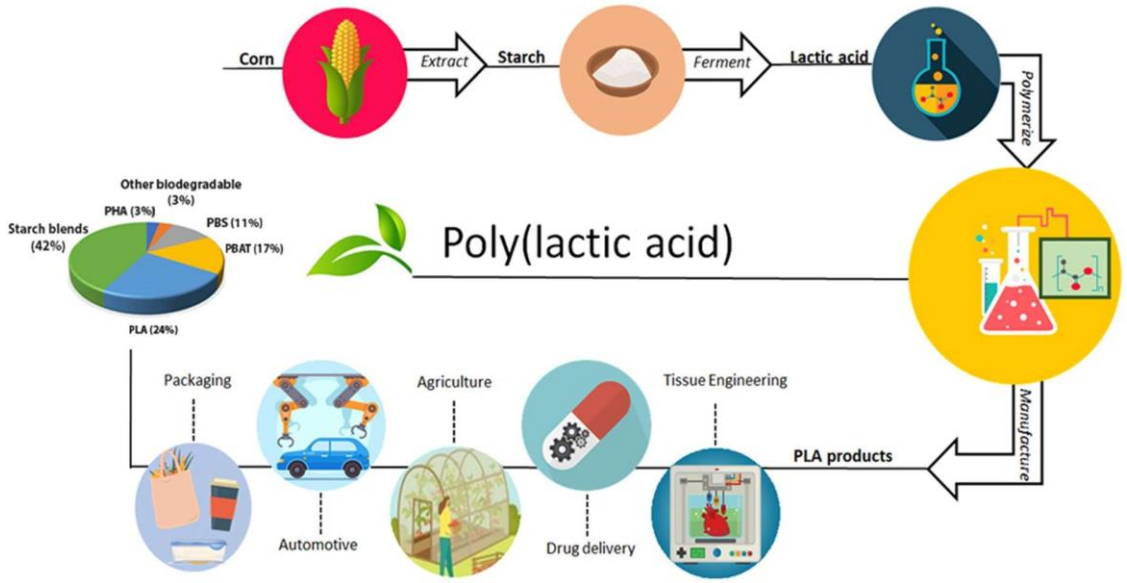
- They are thermoplastic.
- Stable to UV radiation.
- The crystallinity found up to 70%
- Good resistance to moisture.
- Good elasticity properties.

Applications of PHA:

- Packing Industries: Preparation of shopping bags, food containers and cosmetic products. It will reduce the impact of plastic materials.
- Agro products: Synthesis of mulch films, green house films etc
- Textile Industries: Fibers, non-woven fabrics, clothing and carpets
- Medical and Health care: used in the preparation of medical devices, implants due to their bio compatibility. Used in surgical sutures, wound dressing, drug delivery and tissue engineering.
- Toys: Eco friendly and nontoxic kids' toys are prepared with PHA and its derivatives.

2) PLA (Poly Lactic acid)

- PLA is a biodegradable thermoplastic aliphatic polyester derived from animal or plant sources such as cellulose, starch, corn and kitchen waste
- Biobased and biodegradable under industrial composting conditions (at a high temperature, around 58°C).
- Synthesized on a greater scale that is concurrently: biocompatible, biodegradable and biobased.
- Biobased polyester derived from lactic acid (2-hydroxypropionic acid),
- Lactic acid is produced by bacterial fermentation of sugarcane or from the conversion of starch from corn.
- The lactic acid is oligomerized and then catalytically dimerized to make the lactide monomer.
- High molecular weight PLA is produced from the lactide monomer by ring opening polymerization using a stannous octoate catalyst.



Properties of PLA

- PHA polymers are thermoplastic; they become liquid at their melting point.
- UV stable and have a low permeation of water.
- Processability, impact strength and flexibility can be obtained
- PHAs are soluble in solvents such chloroform, dichloromethane or dichloromethane.
- The crystallinity found up to 70%
- Good resistance to moisture.
- Good elasticity properties.

PLA Application in medical field

- PLA used orthopedic device.
- PLA Widely used as a material for drug delivery system.
- PLA used in tissue engineering or regenerative medicine
- PLA used in cardio vascular implants and dental.
- PLA used in skin and tendon healing.

NUCLEIC ACIDS IN VACCINES AND DIAGNOSIS

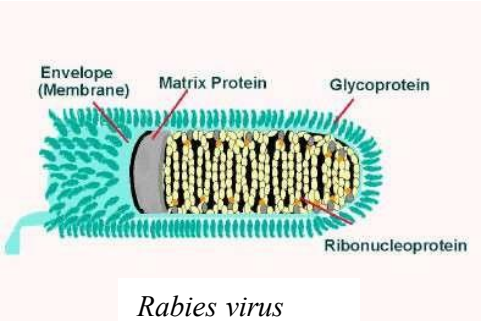
- Nucleic acids are biopolymers, macromolecules, essential to all known forms of life.
- They are composed of nucleotides, which are the monomers made of three components: a 5-carbon sugar, a phosphate group and a nitrogenous base.
- The two main classes of nucleic acids are deoxyribonucleic acid (DNA) and ribonucleic acid (RNA). If the sugar is ribose, the polymer is RNA; if the sugar is the ribose derivative deoxyribose, the polymer is DNA.
- Nucleic acids are naturally occurring chemical compounds that serve as the primary information-carrying molecules in cells and make up the genetic material.

Vaccination:

- Vaccination is a simple, safe, and effective way of protection against harmful diseases, before we come into contact with them.
- Uses your body's natural defenses to build resistance to specific infections and makes your immune system stronger.
- Vaccines train your immune system to create antibodies, just as it does when it's exposed to a disease.

1) DNA vaccine for Rabies

- Preventable viral disease most often transmitted through the bite of a rabid animal.
- Rabies continues to claim upwards of 55,00 human lives each year.
- Most deaths occur in less developed countries in Asia and Africa, affect children <15 years.
- Rabies virus is a highly neurotropic virus that spreads along neural pathway.
- Occur: wild animals- bats, raccoons, skunks, and foxes, although any mammal can get rabies.
- Virus: transmitted through bites or licks by an infected dog

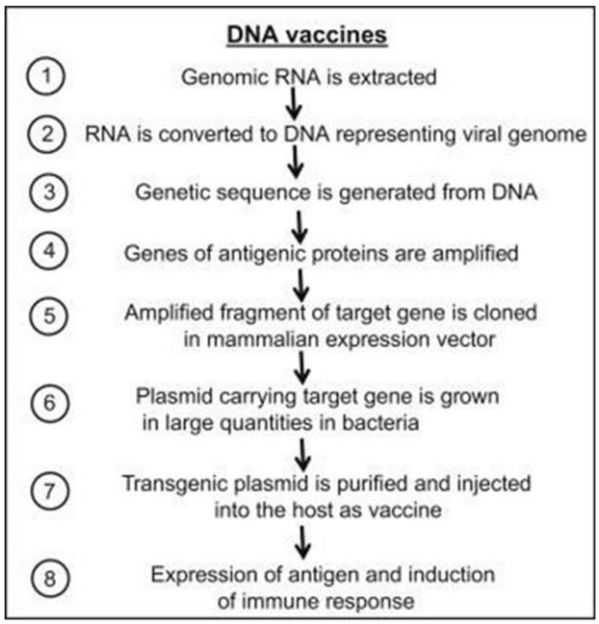


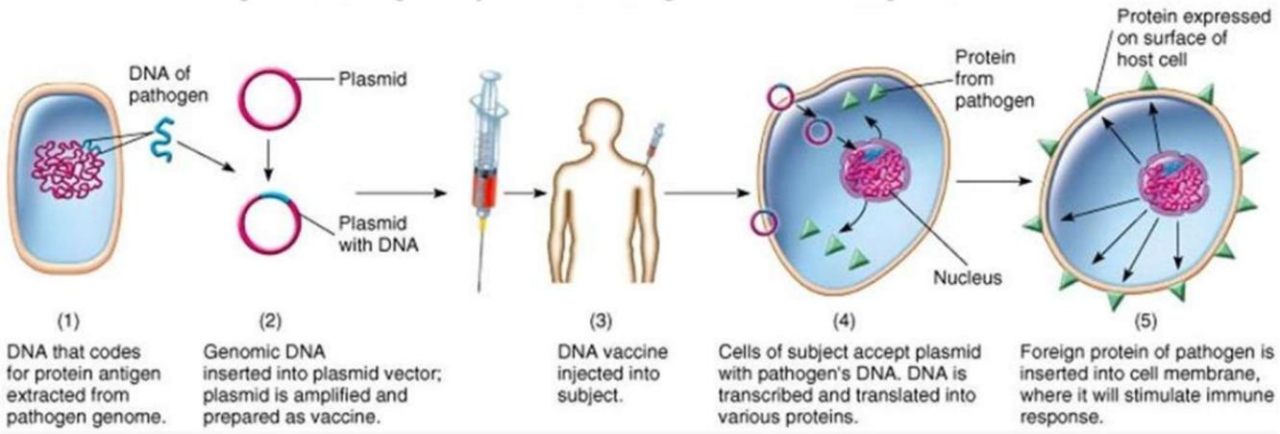
Rabies virus

- The field of DNA vaccination is developing rapidly.
- Consist of synthetic DNA containing the gene that encodes the disease-agent protein.
- The principle behind a DNA vaccine is that the antigen can be expressed directly by host cells in a way that simulates viral infection and invokes an immune response from the host.
- Currently being developed use not only DNA, but also include adjuncts that assist DNA to enter cells, target it towards specific cells, or that may act as adjuvants in stimulating or directing the immune response.

Method of preparation

- The cDNA encoding rabies virus surface glycoprotein or spike protein is amplified in E.coli and cloned into vector to obtain the DRV (DNA Rabies virus) plasmid.
- Plasmid DNA is dissolved in 0.85% NaCl (Saline solution) at a concentration of 1 mg/ml and stored at 4°C until further use.
- The detailed steps involved in DNA vaccine preparation is as follows:





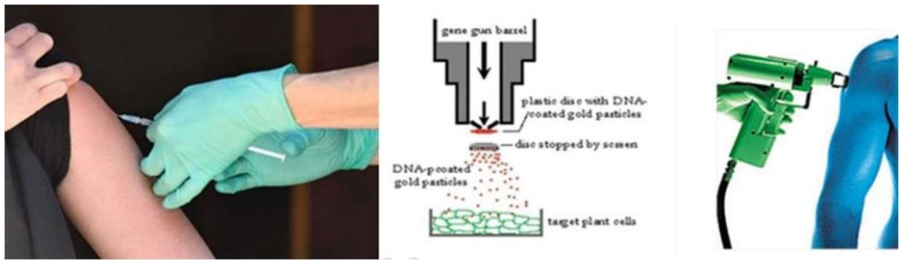
Mode of application:

1) Injection in saline:

- Intramuscularly (IM) in skeletal muscle or intradermally (ID), with DNA being delivered to the extracellular spaces using hypertonic solutions of saline or sucrose.
- Immune responses: affected by; needle type, needle alignment, speed of injection, Volume of injection, muscle type, and age, Sex and physiological condition of the animal being injected.

2) Gene gun delivery:

- Accelerates plasmid DNA (pDNA) that has been adsorbed onto gold or tungsten micro particles into the target cells, using compressed helium as an accelerant.
- Determines the dose of DNA required to raise an effective immune response.



Mode of action:

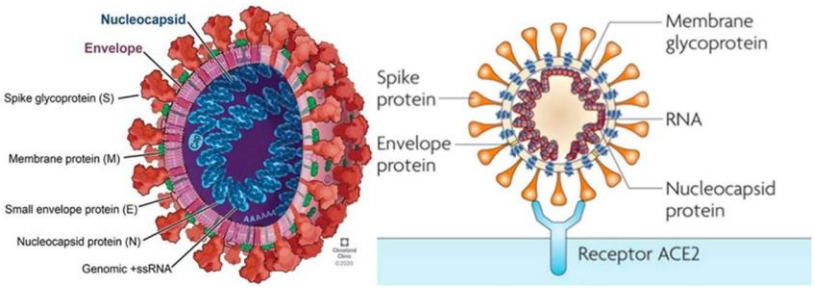
- The vaccine DNA is injected into the cells of the body: The "inner machinery" of the host cells "reads" the DNA and Converts it into pathogenic proteins.
- Because these proteins are recognized as foreign, when they are processed by the host cells and displayed on their surface, implies; the immune system is alerted, which then triggers a range of immune responses

Advantages of DNA vaccine:

- Low capital cost
- For post-exposure treatment, single dose of Rabies DNA vaccine was found to be as potent as 5 dose regimen of cell-culture vaccine in mice
- Reduces cost and facilitates easier product development, compared to conventional vaccines.
- Mass vaccination

2) RNA Vaccine for Covid-19

- COVID-19: caused by a Coronavirus, SARS-CoV-2.
- It is predominantly a respiratory illness that can affect other organs.
- Symptoms may appear 2 to 14days after exposure to the virus.
- Symptoms: fever or chills; cough; shortness of breath; fatigue; muscle or body aches; headache; loss of taste or smell of recent onset; sore throat; congestion or runny nose; nausea or vomiting; diarrhoea.

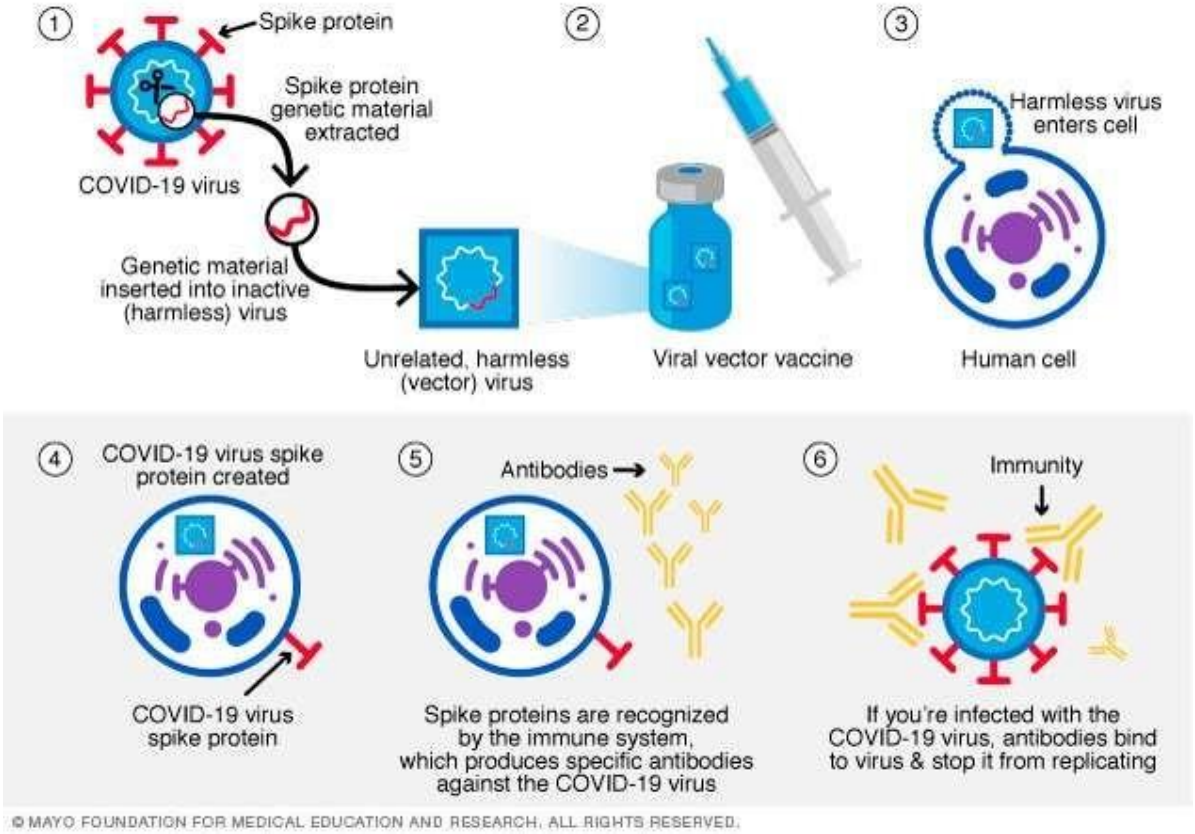


Structure of COVID -19

Messenger RNA (mRNA) vaccine.

- Messenger Ribonucleic acid (mRNA) is molecule that provides cells with instructions for making Protein.
- The mRNA in the vaccine doesn't enter the nucleus of the cell,
- A mRNA vaccine is made using mRNA that gives your cells instructions for how to make the spike protein found on the surface of the COVID-19 virus.

- mRNA vaccines contain the instructions for making the SARS-CoV-2 spike protein. This protein is found on the surface of the virus that cause COVID-19.
- COVID -19 mRNA vaccines are given by injection, usually into muscle of upper arm. After the protein piece is made, the cell breaks down the instruction and gets rid of them.
- After vaccination, the immune cells begin making the spike protein and displays them on cell surfaces.
- This causes the body to create antibodies that can fight the COVID-19 virus.



Advantages of mRNA Vaccine :

- mRNA vaccines as promising alternatives for conventional vaccines mainly lies in the ability to produce the vaccine completely in vivo, along with their high potency, cost-effectiveness, rapid development, and safe delivery
- Lipid nanoparticles (LNPs): commonly used in vivo RNA delivery vectors, protecting the mRNA from enzymatic degradation and facilitating endocytosis and endosomal escape.
- Induce strong mucosal immunity due to their intramuscular administration.

Challenges: Large-scale production, storage, and stability

3) DNA Finger Printing

DNA fingerprinting, also known as DNA profiling or genetic fingerprinting, is a technique used in forensic science to identify individuals by analyzing their unique DNA profiles. Forensic scientists obtain DNA samples from a crime scene, such as blood, semen, or hair follicles, and compare them to DNA samples taken from suspects or victims. The DNA is then extracted from the samples and amplified through a process called polymerase chain reaction (PCR), which makes many copies of the DNA.

The amplified DNA is then subjected to a process called gel electrophoresis, which separates

the DNA fragments according to their size. The resulting pattern of bands, known as a DNA profile or DNA fingerprint, is unique to each individual and can be used to identify a person with a high degree of accuracy. Suspects or victim can be identified by matching the bands of DNA.

Forensic scientists can compare DNA profiles from different samples to determine if they come from the same individual or related individuals. This information can be used to identify suspects or link individuals to a crime scene. DNA fingerprinting has been used to solve numerous high- profile cases and has revolutionized forensic science.

The process of DNA fingerprinting involves several steps:

DNA extraction: DNA is extracted from biological samples such as blood. saliva, or hair found at the crime scene.

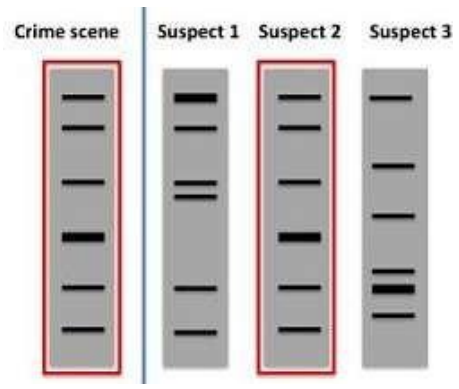
Polymerase chain reaction (PCR): PCR is used to amplify specific regions of the DNA to generate enough material for analysis.

Gel electrophoresis: The amplified DNA fragments are separated by using gel electrophoresis, creating a pattern of bands unique to individual.

Hybridization: A DNA probe is added to the gel, which binds t specific DNA fragment of interest and creates a visible signal.

Analysis: The resulting DNA fingerprint is compared to DNA profiles of potential suspects to determine if there is a match.

DNA fingerprinting has revolutionized forensic science and has become a critical tool in criminal investigations. It is highly accurate and can be used to link suspects to crimes, exonerate innocent individuals, and identify victims of crime or disaster. It is also used in paternity testing and medical diagnosis.



Amplified DNA separated bands

Proteins:

Proteins are large biomolecules that are essential for the structure, function, and regulation of cells, tissues, and organs in living organisms. They are composed of long chains of amino acids that are folded into complex three- dimensional structures.

Proteins have a wide range of functions in the body including:

Structural support: Proteins such as collagen, elastin, and keratin provide strength and elasticity to tissues such

as skin, hair and nails.

Enzymatic catalysis: Enzymes are protein that facilitates chemical reactions in the body by lowering the

activation energy required for the reaction to occur.

Transport: Proteins such as hemoglobin and albumen transport such as myosin nutrients and hormones

throughout the body Immune defence: Antibodies are proteins that recognize and neutral foreign invaders such

as bacteria and viruses.

Regulation: Hormones such as insulin and growth hormone are protein that regulates physiological

processes such as metabolism and growth

Movement: Proteins such as actin and myosin are responsible for contraction of muscle fibers.

Protein as food: Protein is an essential macronutrient that is important for building and repairing tissues

in the body. It is also necessary for the production of enzymes, hormones, and other molecules that are

involved various metabolic processes.

Protein is found in a wide range of foods, including meat, poultry, fi eggs, dairy products, legumes, nuts,

and seeds. Animal-based foods tend to be good sources of complete protein, which means that they

contain all of the essential amino acids that the body needs, while plant-based foods may need to be

combined in specific ways to ensure a complete protein intake

Protein is important for maintaining muscle mass and strength and can help to support weight loss by

promoting feelings of fullness and reducing appetite. It is also important for maintaining bone health and

for supporting the immune system.

Meat protein, on the other hand, is derived from animal sources such as beef, poultry, pork, and fish. Meat is also a complete protein and is a good source of other nutrients such as iron, zinc, and vitamin B12. While both whey protein and meat protein can provide a source of dietary protein, they can have different effects on the body. Whey protein is digested quickly and can help to promote muscle protein synthesis, making it a popular choice for athletes and people looking to build muscle. Meat protein, on the other hand, is generally digested more slowly and may provide a more sustained release of amino acids to support muscle growth and repair. It is important to consume a balanced diet that includes a variety of protein sources, including both plant based and animal-based sources, to ensure adequate intake of all of the essential amino acids and other nutrients.

Whey protein analogues:

Whey protein and meat protein are two different types of dietary proteins.

Whey protein is a type of protein that is derived from milk; specifically the liquid portion that is left over after milk is curdled and strained to make cheese. Whey protein is a complete protein meaning it contains all of the essential amino acids that the body needs. It is commonly used as a dietary supplement and is often consumed by athletes and bodybuilders to support muscle growth and repair.

A whey protein analogue refers to a protein product that mimics the properties of whey protein but is not derived from milk or dairy. This can include plant-based protein sources such as pea protein, soy protein, or rice protein, or even synthetic proteins created in a laboratory.

These analogues are designed to provide similar nutritional benefits as whey protein, such as being a source of High quality protein that is easily digestible and absorbed by the body. They can be used as a dietary supplement for those quality protein that is easily digestible and absorbed by the body. They can be used as a dietary supplement for those who are lactose intolerant, vegan, or have other dietary restrictions that prevent them from consuming whey protein.

An example of a whey protein analogue is pea protein. Pea protein is a plant-based protein source that is derived from yellow split peas. It is a complete protein, meaning it contains all the essential amino acids that the body needs, and has a similar amino acid profile to whey protein. Pea protein is easily digestible and absorbed by the body, making it a popular alternative to whey protein for people who are lactose intolerant, vegan, or have other dietary restrictions. Pea protein is also lower in fat and carbohydrates than whey protein, making it a good choice for those who are looking to manage their calorie intake. Other examples of whey protein analogues include soy protein, rice protein, and hemp protein.

Meat protein analogue:

A meat protein analogue refers to a protein product that mimics the texture, flavour and nutritional properties of meat but is made from non-animal sources. This can include plant-based proteins such as soy, wheat, peas and mushrooms as well as laboratory-grown meat alternatives, also known as cultured meat or cell-based meat.

Plant-based meat alternatives typically use a combination of sources, such as soy and peas, to create a texture and taste that resembles meat. These products are often fortified with vitamins and minerals to provide a nutritional profile that is similar to meat. Additionally they are lower in saturated fat and cholesterol than traditional meat, making them a popular choice for people who are looking to reduce their meat intake for health or environmental reasons.

Cultured meat is produced by taking a small sample of animal cells and growing them in a lab under controlled conditions. The resulting meat is nutritionally identical to traditional meat but is produced without the need for animal slaughter. This technology is still in its early stages but it has the potential to provide a more sustainable and ethical source of meat protein in the future.

Examples of meat protein analogues include products such as Beyond Meat, Impossible Foods, and Quorn, which are plant-based meat alternatives as well as Memphis Meats and Mosa Meat which are companies that are working on production of cultured meat.

Both whey protein analogs and meat protein analogs can provide a sour of dietary protein, while also offering potential health benefits.

Plant based proteins:

Plant-based proteins are proteins that are derived from plants, such as grains, legumes, nuts, seeds, and vegetables. These proteins are a great alternative to animal-based proteins for people who follow a vegetarian or vegan diet, or for people who simply want to incorporate more plant-based Roods into their diet. Some common plant-based proteins include

Legumes: Such as lentils, chickpeas, black beans, kidney beans, and peas.

Nuts and seeds: Such as almonds, peanuts, cashews, chia seeds, flax seeds and pumpkin seeds.

Grains: Such as quinoa, brown rice, oats, and barley.

Vegetables: Such as broccoli, spinach, kale, and Brussels sprouts.

Soy products: Such as tofu, tempeh, and edamame.



Cereals

Plant-based proteins are often lower in saturated fat and cholesterol than animal-based proteins, and they are also rich in fiber, vitamins, and minerals. Additionally, many plant-based proteins are considered "complete" proteins meaning they contain all nine essential amino acids that the body needs to function properly.

Lipids:

Lipids are a group of biomolecules, insoluble in water and soluble in organic solvents. They are an essential component of all living cells, and play a variety of important roles in the body, including providing a source of energy, forming cell membranes, and acting as signalling molecules.

There are several different types of lipids, including: Fatty acids: these are the building blocks of most lipids, and are used to larger molecules such as triglycerides and phospholipids.

Triglycerides: these are the main form of fat stored in the body, and are composed of three fatty acids attached to a glycerol backbone.

Phospholipids: these are a major component of cell membranes, and are composed of two fatty acids attached to a glycerol backbone, along with a phosphate group.

Steroids: These are a type of lipid that includes hormones such as testosterone and estrogens as well as cholesterol, which is an important component of cell membranes.

Lipids are an important source of energy for the body, as they are broken down into fatty acids and used as fuel for cellular processes.

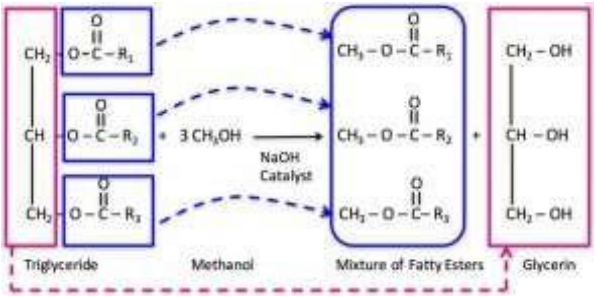
Lipids are source to biodiesel:

Lipids can be a source for biodiesel. Biodiesel is a renewable and friendly alternative to petroleum-based diesel fuel, and can be produced from a variety of sources, including vegetable oils, animal fats, and recycled cooking oils.

The production of biodiesel from lipids involves a process called transesterification, where the lipids are chemically reacted with an alcohol and a case to produce a fatty acid alkyl ester, which is the main component of biodiesel.

Vegetable oils such as soybean oil, canola oil, and palm oil are commonly used as feedstock for biodiesel production as they are rich in triglycerides which can be easily converted into biodiesel through transesterification. Animal fats such as tallow and lard, can also be used as feedstock for biodiesel production.

Using lipids as a source for biodiesel has several benefits, including reducing dependence on fossil fuels, reducing greenhouse gas emissions, and providing an alternative market for agricultural crops.



Transesterification reaction The production of biodiesel typically involves the following steps:

Feedstock preparation: The first step in producing biodiesel is to prepare the feedstock, which is typically vegetable oil, animal fat, or recycled cooking oil. The feedstock must be cleaned and dried to remove impurities and water.

Transesterification: The cleaned and dried feedstock is then reacted with d an alcohol, such as methanol or ethanol, and a catalyst, such as sodium hydroxide or potassium hydroxide, in a reactor vessel.

The reaction breaks down the triglycerides in the feedstock into fatty acid alkyl esters and glycerol. The fatty acid alkyl esters are the main component of biodiesel.

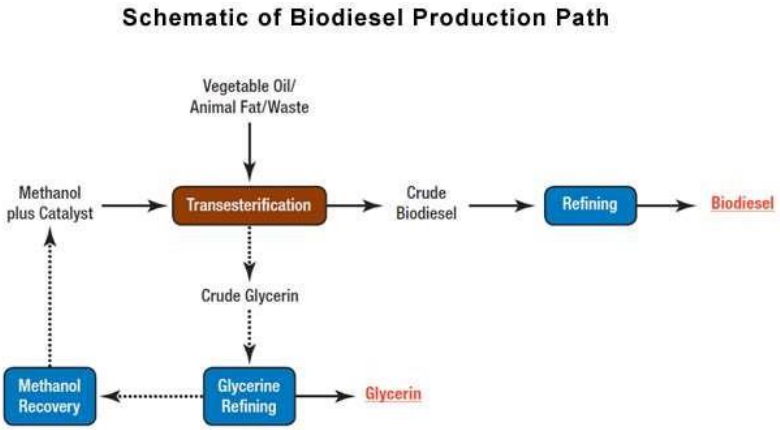
Separation: The fatty acid alkyl esters and glycerol are then separated in a settling tank where the heavier glycerol sinks to the bottom and the lighter biodiesel floats on top. The glycerol can then be removed and s or

used for other purposes.

Washing: The biodiesel is then washed to remove any remaining impurities, such as catalyst residues or unreacted alcohol. This is typically done using water, which is then separated from the biodiesel using s centrifuge or settling tank.

Drying: The final step in producing biodiesel is to dry the fuel to remove any remaining water. This is typically done using a vacuum or a desiccant.

The resulting biodiesel can be used as a fuel in diesel engines, either on its own or blended with petroleum diesel fuel. It is a renewable and environmentally friendly alternative to petroleum- based diesel fuel and can help to reduce dependence on fossil fuels and greenhouse gas emissions.



Flowchart of biodiesel production

Lipids as detergent:

Lipids act as detergents due to their amphipathic nature, Phospholipid is major component of cell membranes. Phospholipids have hydrophilic heads and hydrophobic tails which allow them to form a bilayer in water with the hydrophilic heads facing outwards and the hydrophobic tail the facing inwards. This means they have both hydrophobic (water the repelling) and hydrophilic (water-loving) regions. This allows them to interact with both water and oil making them effective at removing dirt grease and other substances from surfaces.

Lipids can be used in cleaning products such as dish soaps, laundry detergents and all-purpose, one example of a lipid-based cleaning product is soap. Soap is made by mixing fatty acids with an alkali to create a salt. This salt is able to dissolve in and can attach to dirt and grime on surfaces allowing it to be easily removed.

Another example is citrus-based cleaners, which use the natural oils found in citrus fruits to break down and dissolve dirt and grime. These oils contain fatty acids that can act as detergents, allowing them to clean surfaces effectively.

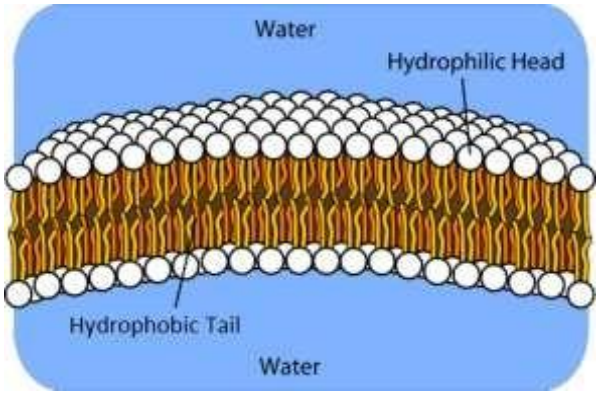


Figure-11 Structure of lipid bi-layers

Enzymes:

Enzymes are proteins that catalyze (speed up) chemical reactions in living organisms that convert substrate to product. They play an essential role in metabolism, which is the set of chemical reactions that occur in cells to maintain life. Enzymes are highly specific in their actions, meaning that they only catalyze specific reactions and are not used up or altered in the process.

Enzymes are made up of long chains of amino acids that fold into complex three-dimensional shapes. The specific sequence of amino acids determines ail the shape of the enzyme and its catalytic activity. Enzymes typically have a pocket or active site where the reaction takes place. This active site is highly specific, allowing it to bind to a particular substrate, or molecule dirt, that the enzyme acts upon.

Enzymes have many important applications in biotechnology, biosensors, medicine, and industry. For example, enzymes are used to produce biofuels, food and beverages, biopharmaceuticals, and other products. They are also used in diagnostic tests and as therapeutic agents for treating disease.

Glucose oxidase:

Glucose oxidase is an enzyme that catalyzes the oxidation of glucose to produce hydrogen peroxide and gluconic acid. It is commonly found in individuals to n many microorganisms, including bacteria and fungi, and is often used in various industrial and biomedical applications.

$$D - Glucose + H_2O + O \rightarrow D - Gluconic\ acid + H_2O_2$$

In the medical field, glucose oxidase is used in diagnostic tests to measure blood glucose levels. Glucose oxidase-based tests are commonly used by people with diabetes to monitor their blood sugar levels. The enzyme reacts with glucose in a blood sample, producing hydrogen peroxide, which is then detected by a colour change or other indicator.

Glucose oxidase as biosensor to detect blood glucose:

Glucose oxidase is commonly used in blood glucose monitoring systems to detect glucose levels in blood. The process involves a biosensor that contains glucose oxidase which catalyzes the reaction of glucose and oxygen to form gluconic acid and hydrogen peroxide.

The biosensor typically consists of a test strip that contains a small amount lignin is highly res of glucose oxidase and other chemicals, including a mediator and the use of plant bio stabilizer. The test strip is inserted into a glucose meter which contains another value-added pr electrode that measures the amount of electrical current generated by the reaction between glucose oxidase and glucose.

When a drop of blood is applied to the test strip the glucose in the blood reacts with glucose oxidase, producing gluconic acid and hydrogen peroxide. The mediator in the test strip facilitates the transfer of electrons from the reaction to the electrode generating an electrical current proportional to the amount of glucose present in the blood. The glucose meter then calculates the glucose concentration in the hood based on the electrical current measured by the electrode. The glucose level displayed on the meter's screen allowing the over to mom their blood e levels and adjusts their insulin dose or diet needed.

Glucose oxidase-based blood glucose monitoring systems are widely d by people with diabetes to manage their condition. These systems provide accurate, convenient, and non-invasive glucose monitoring, allowing individuals to maintain tight control of their blood glucose levels and prevent complications associated with high or low blood sugar.

Lignolytic enzymes:

Lignolytic enzymes are a group of enzymes that are capable of degrading lignin, a complex polymer that is a major component of plant cell walls. Lignin is highly resistant to degradation, which makes it a major barrier to the use of plant biomass as a feedstock for the production of biofuels and other value-added products. There are several types of lignolytic enzymes including

Laccases: Laccases are copper-containing enzymes that catalyze the oxidation of phenolic compounds which are major components of lignin.

Peroxidases: Peroxidases are enzymes that use hydrogen peroxide or other oxidants to oxidize lignin.

Manganese peroxidases: Manganese peroxidases are enzymes that require manganese ions for their activity and are capable of oxidizing both lignin and aromatic compounds

Lignolytic enzymes are produced by a variety of organisms including fungi, bacteria and plants. They play an important role in the natural cycling of carbon in forest ecosystems where they are involved in the decay of woody debris. Lignolytic enzymes are also being investigated for their potential applications in the production of biofuels, bioplastics, and other by-products from lignocellulosic biomass.

Lignolytic enzyme as biobleaching agents:

Lignolytic enzymes have been used in bio bleaching as an eco-friendly alternative to chemical bleaching agents in the pulp and paper industry. The lignin in wood pulp is responsible for the yellowish colour of the paper and it needs to be removed during the bleaching process to produce high quality white paper. Traditional bleaching methods involve the use of harsh chemicals, such as chlorine, which can generate toxic by-products and have negative environmental impacts.

Lignolytic enzymes particularly laccases and peroxidases, have been shown to effectively degrade lignin in wood pulp under mild conditions reducing the need for harsh chemicals in the bleaching process. Laccases and peroxidases are able to catalyze the oxidation of lignin and its derivatives, leading to the breakdown of lignin into smaller, more

soluble compounds that can be removed from the pulp.

Bio bleaching with lignolytic enzymes has several advantages over traditional bleaching methods, including reduced environmental impact, improved paper quality and reduced energy consumption. As a result lignolytic enzymes have the potential to revolutionize the paper industry by providing a more sustainable and cost-effective approach to paper production.